Waves in Two Dimensions

So far, we have only talked about waves in 1 dimension. A wave moves along a string, or a planar sound wave moves in a single direction. Things got a bit more complicated when we added in a change in medium/wave speed. The behavior of waves gets *vastly* more complicated and/or rich and interesting when you look at them in two dimensions. Even though the behavior is a lot more complicated, we'll still get the same behavior for sound, light, and water waves. It remains true that if you've seen one wave, you've seen them all.

These wave effects can matter tremendously in music. Why is vibrato so widely used? What happens if you wire a speaker backwards? Looking at the properties of 2-D waves can help us understand these questions.

1. Single Source

Picture a pebble tossed into a pond. You get circular ripples moving outwards. The amplitude of the wave does decrease with distance, but we'll ignore that for now. If we have a single source, then the wave behavior will look like

$$f(r - vt) \tag{1}$$

Equation 1 looks very much like the wave equations we're used to $f(x - vt)$, but we've replaced x with r where r is the distance from the soure to a given point in 2-dimensional (or higher!) space. Because every point in space with the same value of r will behave the same way, and points with constant distance from a source form circles, our wave will look like circular patterns moving away from the source. If we do want to worry about waves weakening with distance, then Equation 1 turns into

$$\frac{r^\alpha}{f}(r - vt) \tag{2}$$

where α depends on how many dimensions the wave is moving in. For 3D, $\alpha = -1$, that is, wave amplitude/pressure fall off as $\frac{1}{r}$.

Unfortunately Desmos is not great at the sort of 2D plots we'll want to look at here, but you're welcome to look at the python script waves_2d.py in the python directory on github. That will make 2D animations showing waves spreading outwards from a variety of point setups. If you uncomment the line
pos=[[0,0]]
the script will show a sine wave moving away from a source at $[0, 0]$. I haven't worried about the amplitude decay; as we'll see, our hands will be full enough without that (and the amplitude usually doesn't matter much anyways).

2. Two Sources, In-Phase

If we have two sources, the behavior gets a lot more complicated. Again, picture a pond but this time you threw in two pebbles. Each pebble makes a set of rings, but now some places the rings

reinforce each other, and some places they cancel out. If we ask "what happens at point (x, y) when we have a source of waves at (x_1, y_1)?", we first need to note that

$$r = \sqrt{(x - x_1)^2 + (y - y_1)^2} \tag{3}$$

Once we have calculated r, which does *not* change with time, then we can go look up our wave height for any time t by looking at Equation 1. If we have a sine wave, then we have our usual situation

$$\sin(kr - \omega t) \tag{4}$$

where as usual the wave speed $v = \frac{omega}{k}$. Plugging in Equation 3 for r into Equation 4 will let us work out the height of our wave given a point (x, y).

What happens when we add a second source? The wave equation (at least in the cases we'll consider) stays linear. That is, we just add the two waves together. If our sources are at (x_1, y_1) and (x_2, y_2) and r_1 and r_2 are the distances from our chosen point (x, y) and the two sources, our sine wave now becomes

$$\sin(kr_1 - \omega t) + \sin(kr_2 - \omega t) \tag{5}$$

To try and make more sense of this, we'll introduce a transform. Let $r = \frac{r_1 + r_2}{2}$ and $dr = \frac{r_1 - r_2}{2}$. In words, r is the distance from our point to the *center* of the two sources, and dr is the difference in distance from one point relative to the center. If we add the two equations together we get

$$r + dr = r_1/2 + r_1/2 = r1 \tag{6}$$

and

$$r - dr = \frac{r_1 + r_2}{2} - \frac{r_1 - r_2}{2} = r_2 \tag{7}$$

We can plug Equations 6 and 7 into Equation 5 to get:

$$\sin(k(r + dr) - \omega t) + \sin(k(r - dr) - \omega t) \tag{8}$$

We're now going to use $\sin(a + b) = \sin(a)\cos(b) + \cos(a)\sin(b)$ with $b = dr$ to get

$$\sin(kr - \omega t + kdr) + \sin(kr - \omega t - kdr) \tag{9}$$

$$= \sin(kr - \omega t)\cos(kdr) + \cos(kr - \omega t)\sin(kdr) + \sin(kr - \omega t)\cos(-kdr) + \cos(kr - \omega t)\sin(-kdr) \tag{10}$$

$$= 2\sin(kr - \omega t)\cos(kdr) \tag{11}$$

Equation 11 should look rather familiar. Mathematically, it's the same as the beating from two closely-spaced frequencies, but the interpretation is now rather different. The first term $\sin(kr - \omega t)$ is just a sine wave coming from the midpoint of the two sources.

The second term in Equation 11 is an overall amplitude. If we pick dr to be very small - that is, we're at a point in space where the *difference* in distance between our point an the two sources is very small, then kdr is very small, and $\cos(kdr)$ is very close to one. If we're at a point that is the same distance from our two sources, then their waves *add* and we get a *strong* signal. This is called *constructive interference* - the two waves are adding *in-phase*. What happens if $kdr = \pi/2$? Now $\cos(kdr) = 0$, and the waves are cancelling each other out. We don't get a signal, and this is

called *destructive interference*. If we convert from k to wavelength λ then we see we get destructive interference at $dr = \lambda/4$. Recall that we defined dr to be the change in distance between the midpoint and each source, so the difference in distance is $\lambda/2$. That is, if we are exactly half a wavelength further from one source than another, they destructively interfere.

We get constructive interference not just for $dr = 0$. Whenever the distance difference is an integer number of wavelengths we'll get constructive interference.

3. Pattern from Two Sources

Let's consider two wave sources. We'll put the first at the origin $(0, 0)$ and the second at a distance d along the $y-$axis, so its position is $(0, d)$. We want to find the distances from the two sources to a receiver that's a distance r from the origin that makes an angle θ with the $x-$axis. The position of the receiver is then $(r\cos(\theta), r\sin(\theta))$. The distance from the first source to the receiver is obviously r (after all, that's why we put the first source at the origin). Now we want to work out the distance from the receiver to the second source. Recall that the distance between two points separated by $(\delta x, \delta y)$ is $\sqrt{\delta x^2 + \delta y^2}$. The $x-$separation is just $r\cos(\theta)$ since source 2 is at $x = 0$, but the $y-$separation is $r\sin(\theta) - d$. We then get the distance between source 2 and the receiver r_2 is:

$$r_2 = \sqrt{(r\cos(\theta))^2 + (r\sin(\theta) - d)^2} \tag{12}$$

We can expand the second term in the equation since $(a - b)^2 = a^2 - 2ab + b^2$ to get:

$$r_2 = \sqrt{r^2\cos(\theta)^2 + r^2\sin(\theta)^2 - 2r\sin(\theta)d + d^2} \tag{13}$$

Since $\cos^2 + \sin^2 = 1$ we can combine the first terms to get

$$r_2 = \sqrt{r^2 - 2r\sin(\theta)d + d^2} \tag{14}$$

We can pull a factor of r^2 out of the square root by multiplying/dividing the second term by r and the third term by r^2 to get:

$$r_2 = \sqrt{r^2\left(1 - 2\frac{\sin(\theta)d}{r} + \frac{d^2}{r^2}\right)} \tag{15}$$

$$r_2 = r\sqrt{1 - 2\frac{\sin(\theta)d}{r} + \frac{d^2}{r^2}} \tag{16}$$

This expression is exactly correct, but can be awkward to work with. We can make some progress if we assume we're a long ways from the sources, so that r is much, much larger than d. If that's the case, then the second term, which has a factor of $\frac{d}{r}$ will be a lot smaller than 1, and the third term, which is $\frac{d^2}{r^2}$ will not only be a lot smaller than 1, it will also be a lot smaller than the second term.

At this point, we'll want to know what $\sqrt{1 + \delta}$ is approximately equal to when δ is very small. We can figure this out by looking at $(1 + \epsilon)^2$ where ϵ is also a very small number. We know that

$(1+\epsilon)^2 = 1+2\epsilon+\epsilon^2$, but if ϵ is small, then ϵ^2 is tiny. For instance, if $\epsilon = 0.001$, then $\epsilon^2 = 0.000001$. If we throw away the extremely tiny ϵ^2 term, then we have

$$(1+\epsilon)^2 \simeq 1+2\epsilon \tag{17}$$

Now we take a square root of both sides and we have

$$1+\epsilon \simeq \sqrt{1+2\epsilon} \tag{18}$$

Finally, we define $\delta = 2\epsilon$, and after swapping the left and right sides of the equation, we get

$$\sqrt{1+\delta} \simeq 1+\frac{\delta}{2} \tag{19}$$

This gives us the very useful result that the square root of 1 plus a small number is 1 plus half of that small number. We can apply this to Equation 16 to get the approximate distance when r is much larger than d:

$$r_2 \simeq r\left(1-\frac{d\sin(\theta)}{r}\right) = r - d\sin(\theta) \tag{20}$$

The *difference* in distances is

$$r - r_2 = r - (r - d\sin(\theta)) = d\sin(\theta) \tag{21}$$

Equation 21 tells us something remarkably simple: if we are a long ways from the sources, the difference in distance between the two sources and our receiver is just $d\sin(\theta)$ where again d is the distance between the two sources, and θ is the angle our receiver makes with the line from the first source that is perpendicular to the line connecting the two sources.

4. Implications of Delay

Equation 21 goes much deeper than just a pair of speakers. In general, if we try to squeeze a wave through a hole of diameter d, it will spread out afterwards by roughly an angle of $\sin(\theta) = \frac{\lambda}{d}$. This is true for light, for water, for sound. Through the magic of quantum mechanics, it's even true for electrons and atoms. The exact pattern will depend in detail on the shape of the hole (or holes), but the basic idea that the waves will spread out over an angle of roughly $\frac{\lambda}{d}$ is very general.

If we send a bunch of different waves through the same hole, the longer waves will spread out more. You've probably noticed this walking by an open door/window with music playing inside. The bass spreads out, so you hear that when you're a ways from the door. Without high notes, the music will sound muffled. As you get closer, the treble starts to come in. When you're right in front of the door then the music will sound normal, as if the rest of the wall isn't even there.

We can also get an idea what happens if there are obstructions. If we have a wave coming in, and an obstruction blocks some of it, what happens? There's no sound coming through the blockage. One way we can mimic this by putting in a bunch of tiny speaker that are perfectly out of phase with the incoming signal. Everywhere there's a speaker, it cancels the sound perfectly. The sound from the speakers will spread out, and we can add that spreading to the original wave

to figure out what we'll actually hear. If the hole is smaller than a wavelength, the sound spreads out in all directions. There isn't very much of it, though, so the overall sound level doesn't really change, and doesn't really depend on position. Blockages smaller than a wavelength may as well not be there.

If we're wider than a wavelength, though, we're going to cancel the wave behind the blockage. How far away will the sound still be blocked? If you think about it for a moment, you might be able to convince yourself that once the sound from the cancelling speakers spreads out by more than the width of the blockage, the blocking starts to go away. If we let θ be small, then the angle we spread over is $\theta = \frac{\lambda}{d}$. If we're at a distance r, the physical size we've spread over is $r\theta$. The blockage fades out when that width becomes bigger than the blockage, so $r\theta = d$ or

$$r\frac{lambda}{d} = d \tag{22}$$

$$r = \frac{d^2}{\lambda} \tag{23}$$

Equation 23 tells you something possibly very useful. If you have something of width d that blocks sound, if you go a distance much larger than $\frac{d^2}{\lambda}$ away, then you won't notice the block. Let's say you're in a concert sitting behind a pillar. Let's say the upper frequency we'll care about is 10 kHz (it's only very young people who hear up to 20 kHz), or about 3cm wavelength. How far do you want to be away from a pillar? If the pillar is say 1m wide, then our distance would be $\frac{1^2}{0.03} = 30m$. That's pretty far! If the pillar is a more modest 0.5m wide, and you care about 1 kHz/30cm wavelength, then the distance is a much shorter $\frac{0.5^2}{0.3} = 0.83m$.

5. Localization of Sounds

The same principles apply if we're trying to figure out where a sound is coming from. Our ears are very, very good at measuring the time delay between when a sound hits your ears. Our brain turns that into a direction. The math is identical if we have two sources and one receiver, or one source and two receivers. If we have a pure sine wave coming in, though, our ears aren't sure what to do. There's no way to label peaks of a sine wave, so the best our ears can do is make sure to line up peaks in our left and right ear. Depending on the frequency, there are a bunch of possible angles/directions that will do that. There's no way to know which is the right direction, and so we have a really, really hard time telling where pure sounds are coming from. Those directions change as we change frequency, though, so a sound that has lots of different frequencies in it is much easier to localize.

6. Beaming and Noise Rejection

Another example of the power of delay is beaming of sound, or its equivalent of noise rejection. Let's say you have a bunch of speakers and only want to play sound in one direction. You know the music you want to play in advance, and can decide *when* each speaker will play. If you want to play in a direction θ, you add a (different) delay to each speaker to exactly cancel the path length

difference. The more speakers you have, the better you can do this, where better means sending sound only in that direction.

Running in reverse, you might also want to pick up sound only in a given direction (think trying to hear a soloist in a noisy room). If you have a bunch of microphones and sync them up, you can add them so that the soloist is in phase. Noises coming from elsewhere won't line up across the microphones, so if you add the signals together, you'll cancel the noise from elsewhere. One power of this (rather than say using a directional microphone) is that you could pick several directions. If you record all your microphones separately, you could even do this in post-production.

7. Huygens Principle and Gratings

One final point that I'll make is what's called *Huygens' Principle*. Huygens' Principle states that if I have a wave, I can treat every point on that wave as the source of a new wave moving out spherically. If you think about it the right way it can seem rather obvious, but it can also be very hard to wrap one's mind around. It can be even harder to apply mathematically.

The obvious way of thinking is "well, I know that the sum of two solutions of the wave equation is also a solution to the wave equation. If I take a snapshot of my setup, I can take a single tiny patch and set the rest of the setup to zero. That tiny patch will just act like a wave moving out circularly. Of course the tiny patch next to my first patch also started out with some position, so I'll add another solution centered on this tiny patch." Repeat adding tiny patches until you match the wave you started with. Each one of those tiny patches makes a wave centered on that patch. The sum of all those circular waves must then give us the behavior of our original system.

The math for how this all works out is *fiercely* complicated so we won't be doing it in general. Qualitatively, though, how does a ripple in a pond work? It sure doesn't look like every point on the ripple is sending out its own ripple. However, it is. It's just that those waves cancel out precisely almost everywhere, and only add constructively where the original wave is heading.

Given a point in space to observe, we do know where these tiny waves are coming from and how far they take to get to our point. If we pick one point on the wave as a source, we know the phase between that point and our observer. What if we took a bunch of tiny walls, and blocked all the bits of the wave that are out of phase at our observer, but let all the bits that are in phase through? Our observer would see a big signal. Just by judicious blocking of the wave, we can send a beam in an entirely different direction. When we do this with light it's called a *diffraction grating*. We saw one in class where the laser light spread out. We could do the same thing with sound. If you were to listen to a sound through a wooden fence with wide planks and narrow gaps, different pitches would be bent by different amounts. If someone behind the fence played a chord, depending on your position relative to the fence, you might only hear one note, and as you move, the note you hear will change!

If you've seen one wave, you've seen them all...